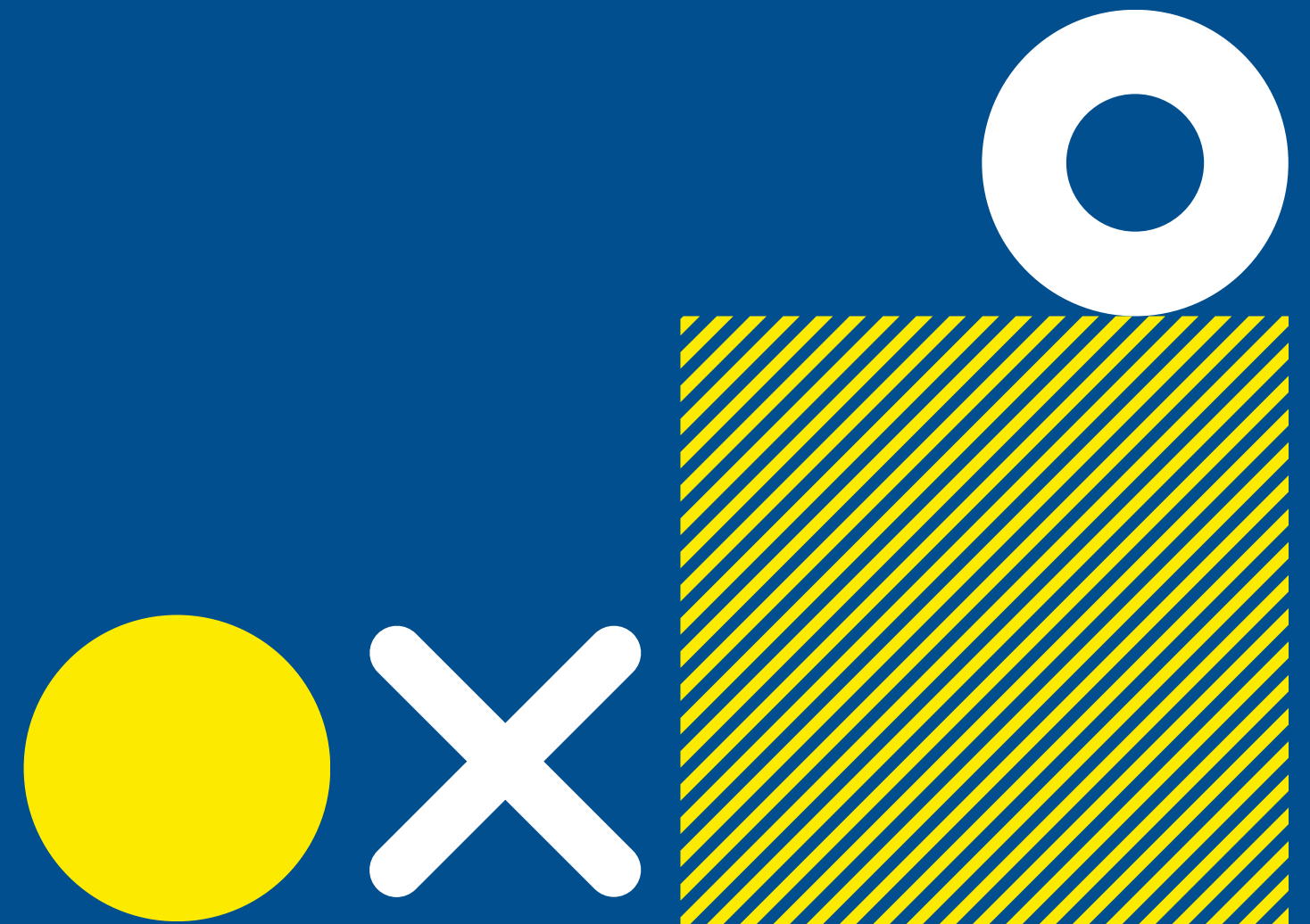
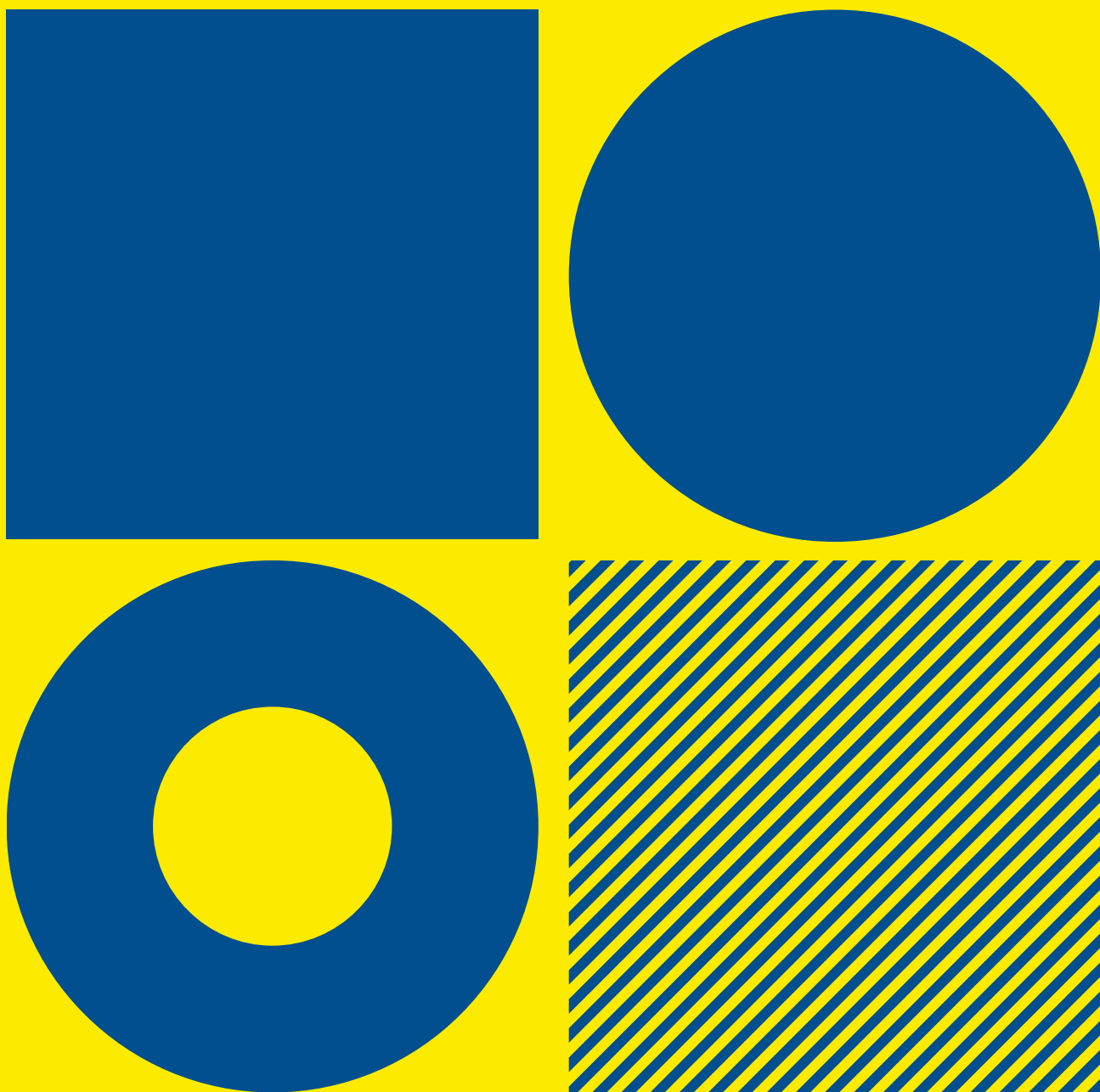


Online Retail Store Analysis



By: Jason Joseph Tjiadi





Background





The dataset contains all transactions for UK-based and registered online retail shops between December 2009 and December 2011 that mainly focuses on gift-ware

Background

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 7:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 7:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 7:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 7:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 7:45:00	1.25	13085.0	United Kingdom

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1067371 entries, 0 to 1067370
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Invoice      1067371 non-null object
1   StockCode    1067371 non-null object
2   Description  1062989 non-null object
3   Quantity     1067371 non-null int64
4   InvoiceDate   1067371 non-null datetime64[ns]
5   Price        1067371 non-null float64
6   Customer ID  824364 non-null float64
7   Country      1067371 non-null object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 65.1+ MB
```

The dataset contains 1,067,371 rows and 8 columns with attribute information as follows:

- Invoice : Invoice number uniquely assigned to each transaction.
- StockCode: Product code uniquely assigned to each distinct product.
- Description: Product name
- Quantity: The quantities of each product (item) per transaction.
- InvoiceDate: The day and time when a transaction was generated.
- UnitPrice: Product price per unit in sterling (£).
- CustomerID: Customer number uniquely assigned to each customer.
- Country: The name of the country where a customer resides.

Pre-Processing

```
#Check percentage of duplicated values
df.duplicated().sum()/len(df)
```

```
np.float64(0.032167821685243465)
```

```
duplicate = df[df.duplicated()]
```

```
print("Duplicate Rows :")
```

```
# Print the resultant Dataframe
duplicate.head()
```

```
#No treatment needed
```

```
Duplicate Rows :
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
371	489517	21912	VINTAGE SNAKES & LADDERS	1	2009-12-01 11:34:00	3.75	16329.0	United Kingdom
383	489517	22130	PARTY CONE CHRISTMAS DECORATION	6	2009-12-01 11:34:00	0.85	16329.0	United Kingdom
384	489517	22319	HAIRCLIPS FORTIES FABRIC ASSORTED	12	2009-12-01 11:34:00	0.65	16329.0	United Kingdom
385	489517	21913	VINTAGE SEASIDE JIGSAW PUZZLES	1	2009-12-01 11:34:00	3.75	16329.0	United Kingdom
386	489517	21821	GLITTER STAR GARLAND WITH BELLS	1	2009-12-01 11:34:00	3.75	16329.0	United Kingdom

```
#Check percentage of missing value
df.isnull().sum()/len(df)
```

```
0
```

Invoice	0.000000
StockCode	0.000000
Description	0.004105
Quantity	0.000000
InvoiceDate	0.000000
Price	0.000000
Customer ID	0.227669
Country	0.000000

```
dtype: float64
```

In this stage, the data is being checked for duplicated values as well as missing values. Since about 22% of the CustomerID are missing and that customer data is pretty crucial for the analysis, therefore we will remove rows with missing values. Here we also make sure that all other data points are valid and usable with one of our treatments filtering quantity sold greater than 0.

[Code](#)

Pre-Processing

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01	6.75	13085.0	United Kingdom
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4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01	1.25	13085.0	United Kingdom

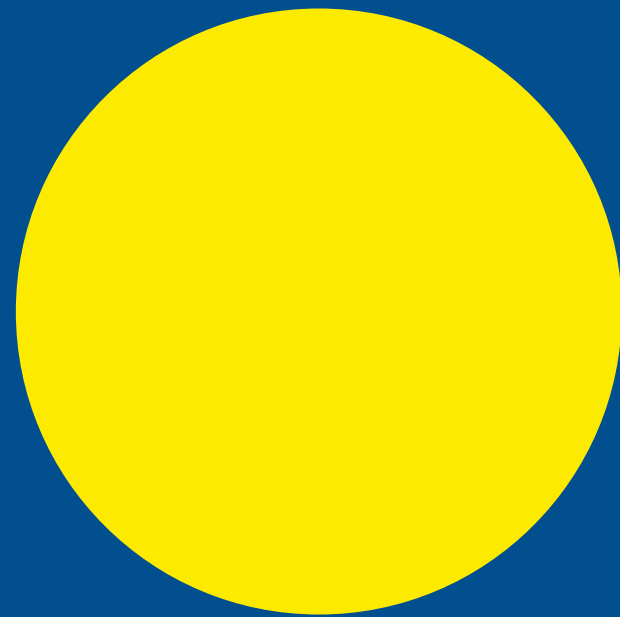
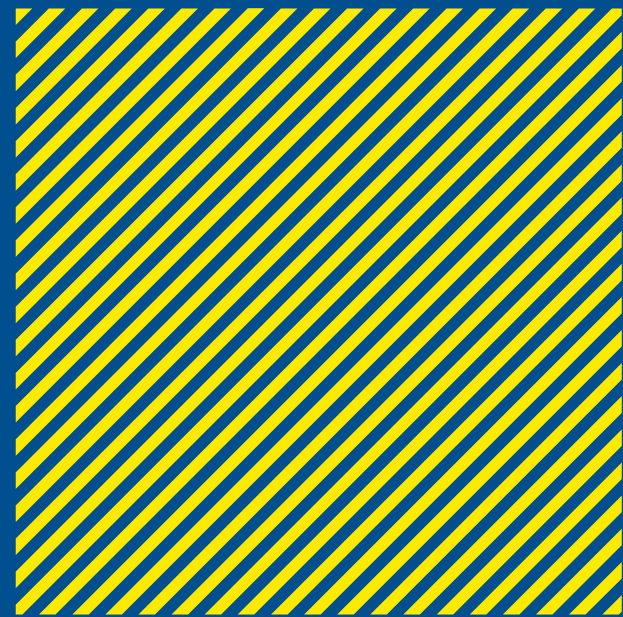
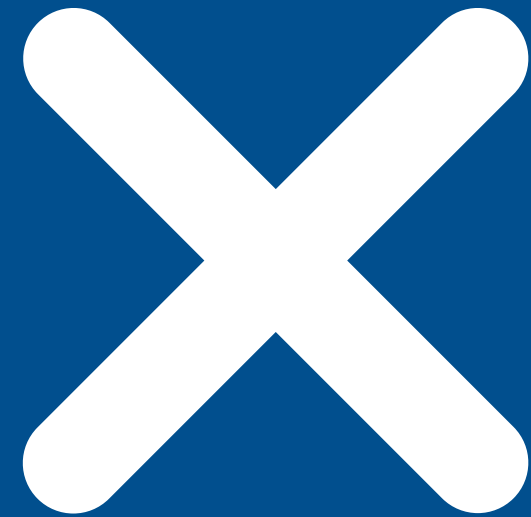
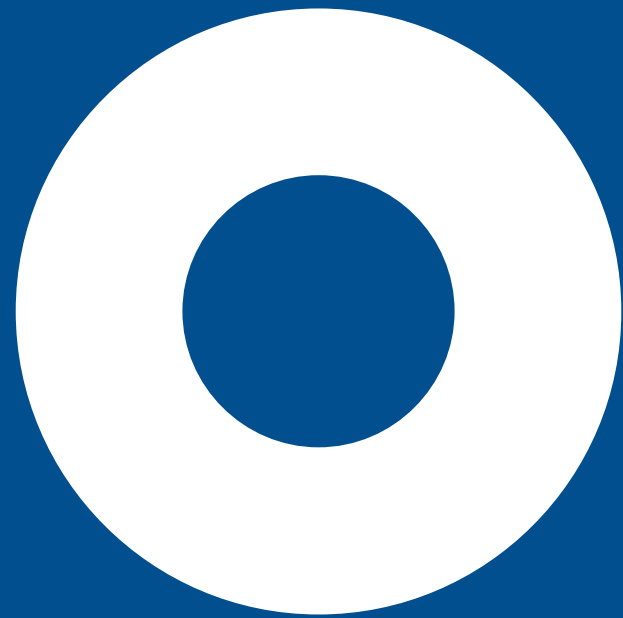
```
#Check percentage of missing value
df.isnull().sum()/len(df)*100
```

	0
Invoice	0.0
StockCode	0.0
Description	0.0
Quantity	0.0
InvoiceDate	0.0
Price	0.0
Customer ID	0.0
Country	0.0

```
<class 'pandas.core.frame.DataFrame'>
Index: 824364 entries, 0 to 1067370
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Invoice          824364 non-null object
1   StockCode       824364 non-null object
2   Description      824364 non-null object
3   Quantity        824364 non-null int64
4   InvoiceDate      824364 non-null datetime64[ns]
5   Price           824364 non-null float64
6   Customer ID     824364 non-null object
7   Country         824364 non-null object
dtypes: datetime64[ns](1), float64(1), int64(1), object(5)
memory usage: 56.6+ MB
```

This is what the data looks like after cleansing.

We end up with 824,364 rows and 8 columns.



*Problem
Statement*



Problem 1

Investigate consumers' transactional patterns and behaviour

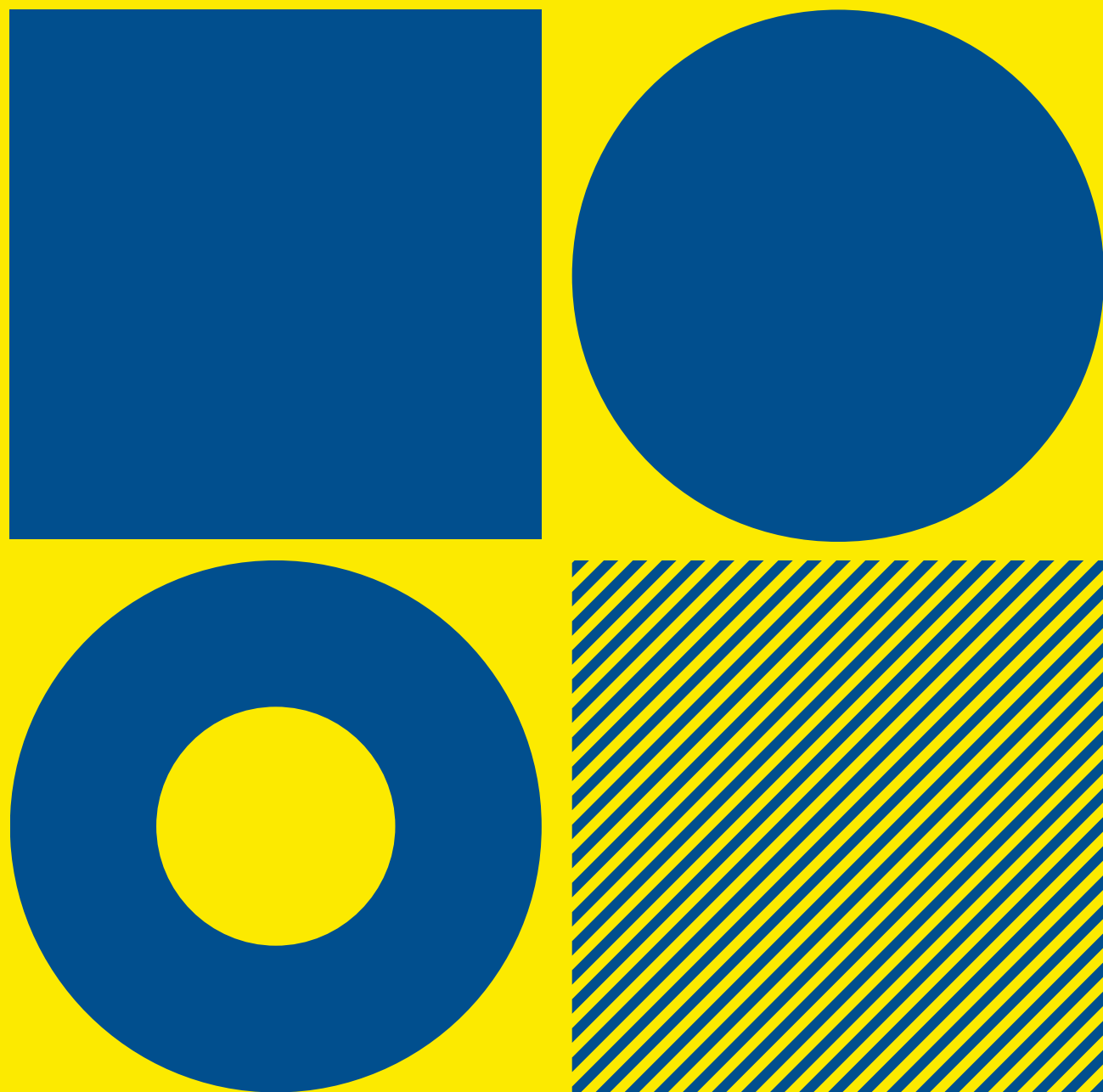
Problem 2

Identify the best selling product within the store

Problem 3

Figure out activation strategy to retain existing customers





Methodology





RFM Analysis

Evaluate the customer's profile and demography to come up with a better marketing strategy

Dashboard

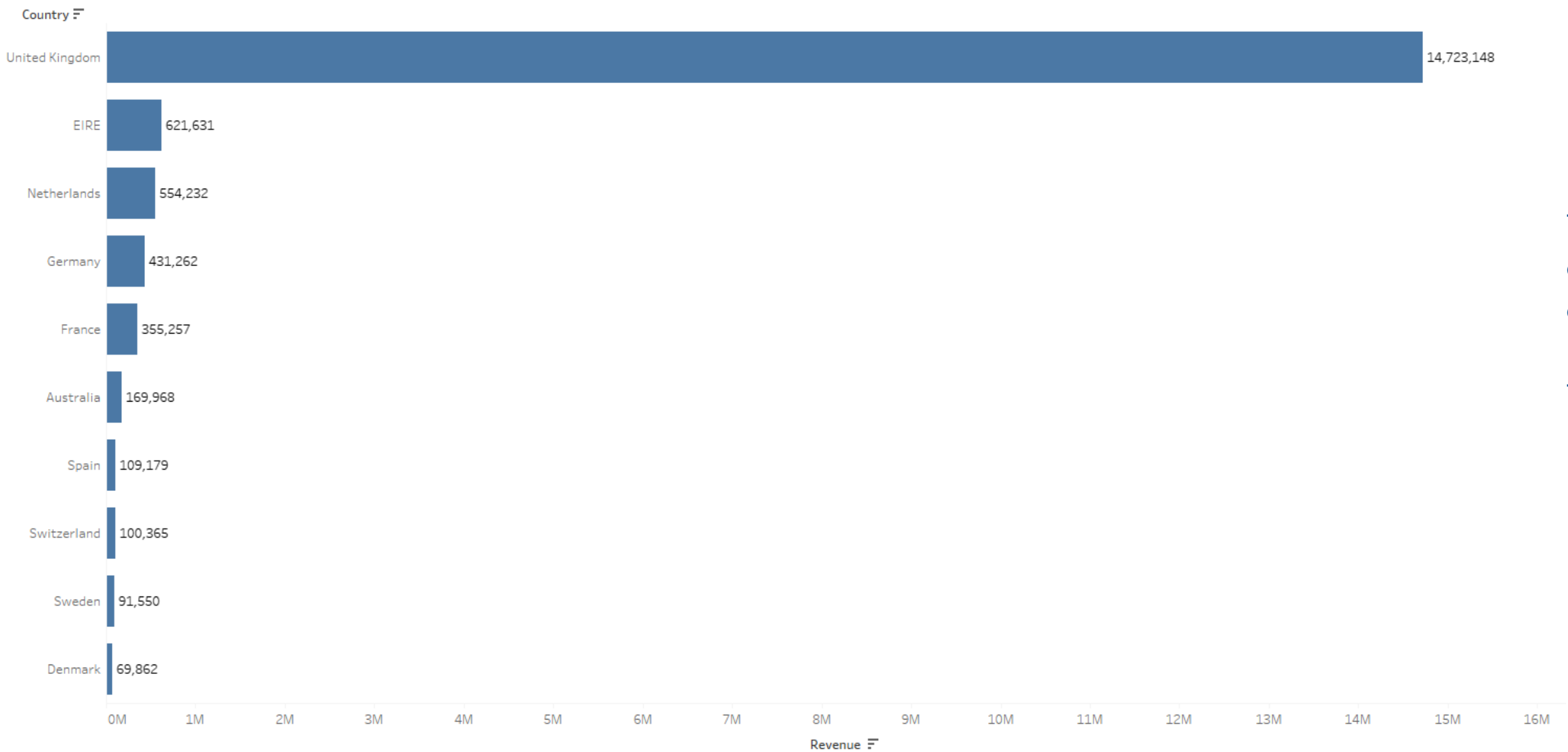
Understanding customer behaviour and patterns

Cohort Analysis

Measure how frequent customers return to make another purchase

Regional Markets

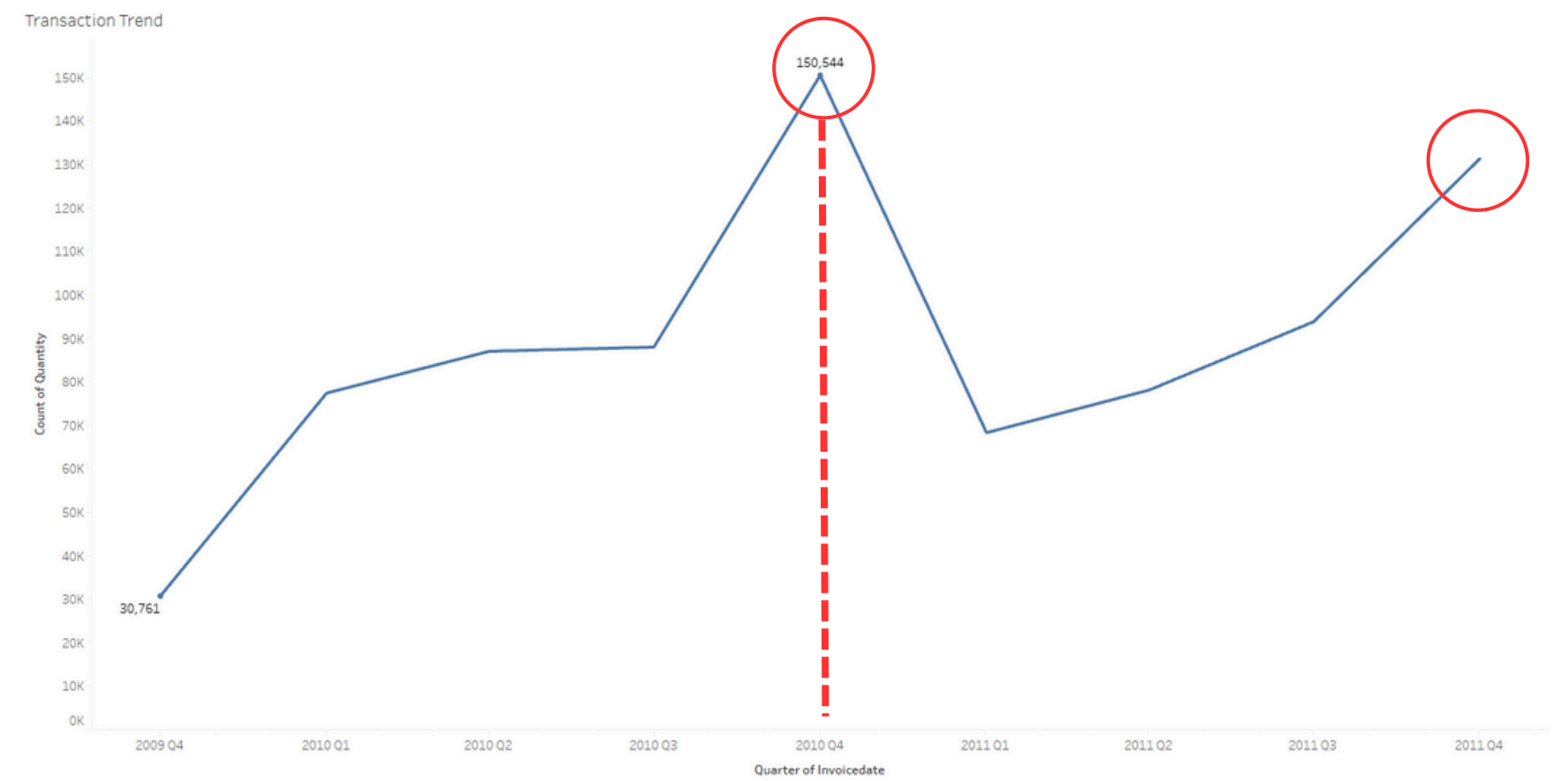
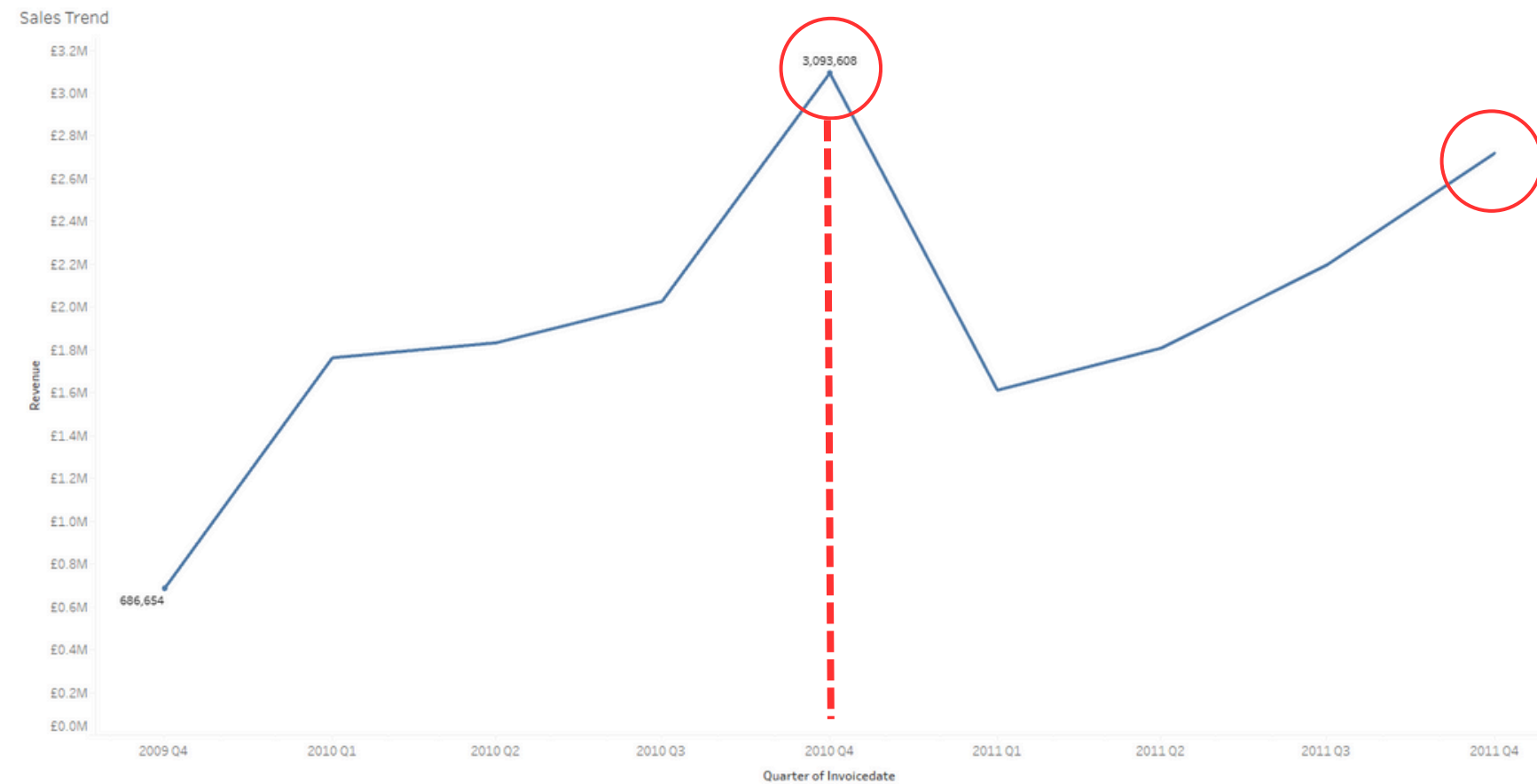
Customer's Geographic Contribution



The UK Market still dominates most of the customers who have purchased from the online retail shop.

This indicates a strong presence within the local market

Time Series Trend

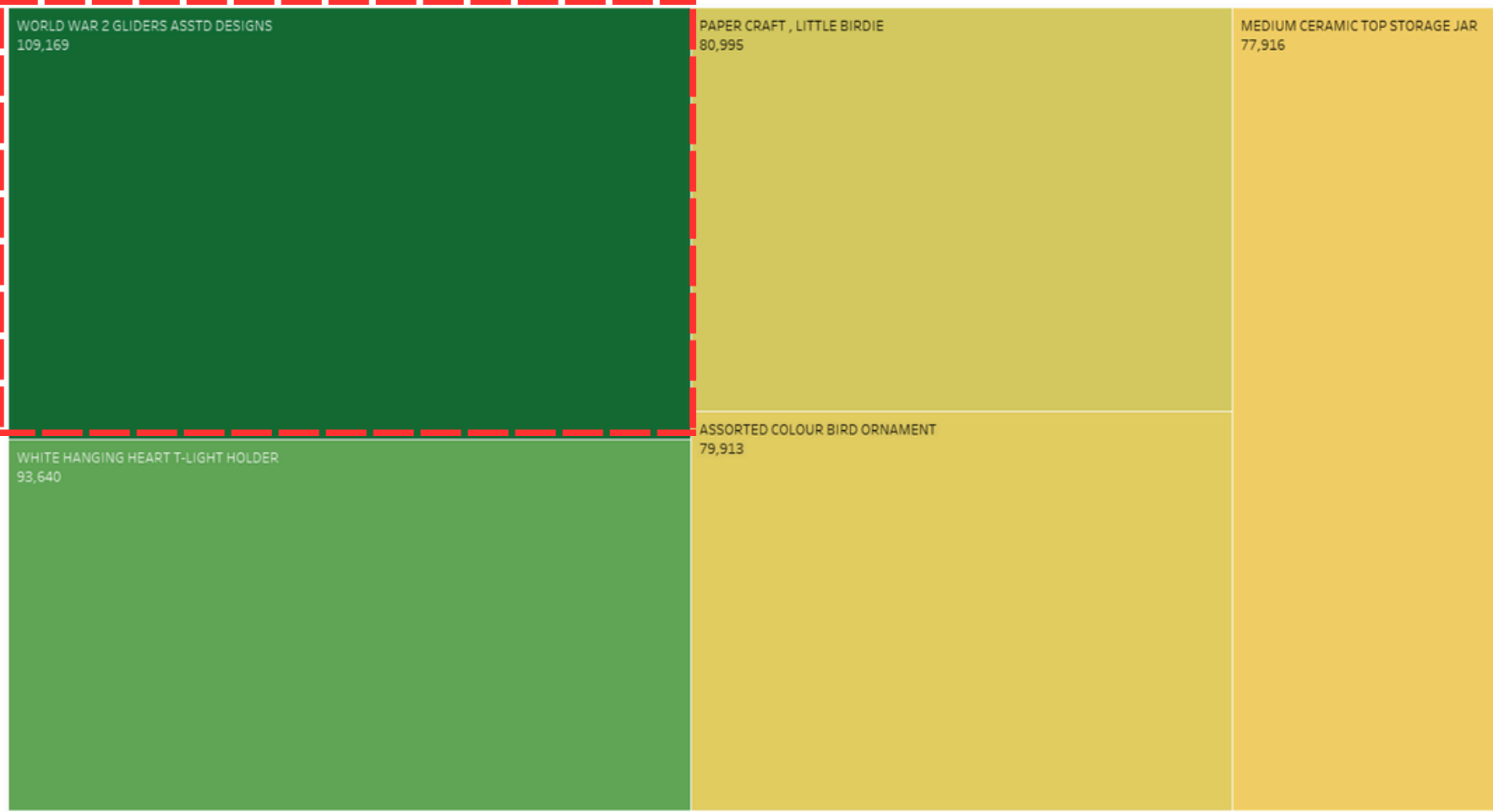


Key Insights

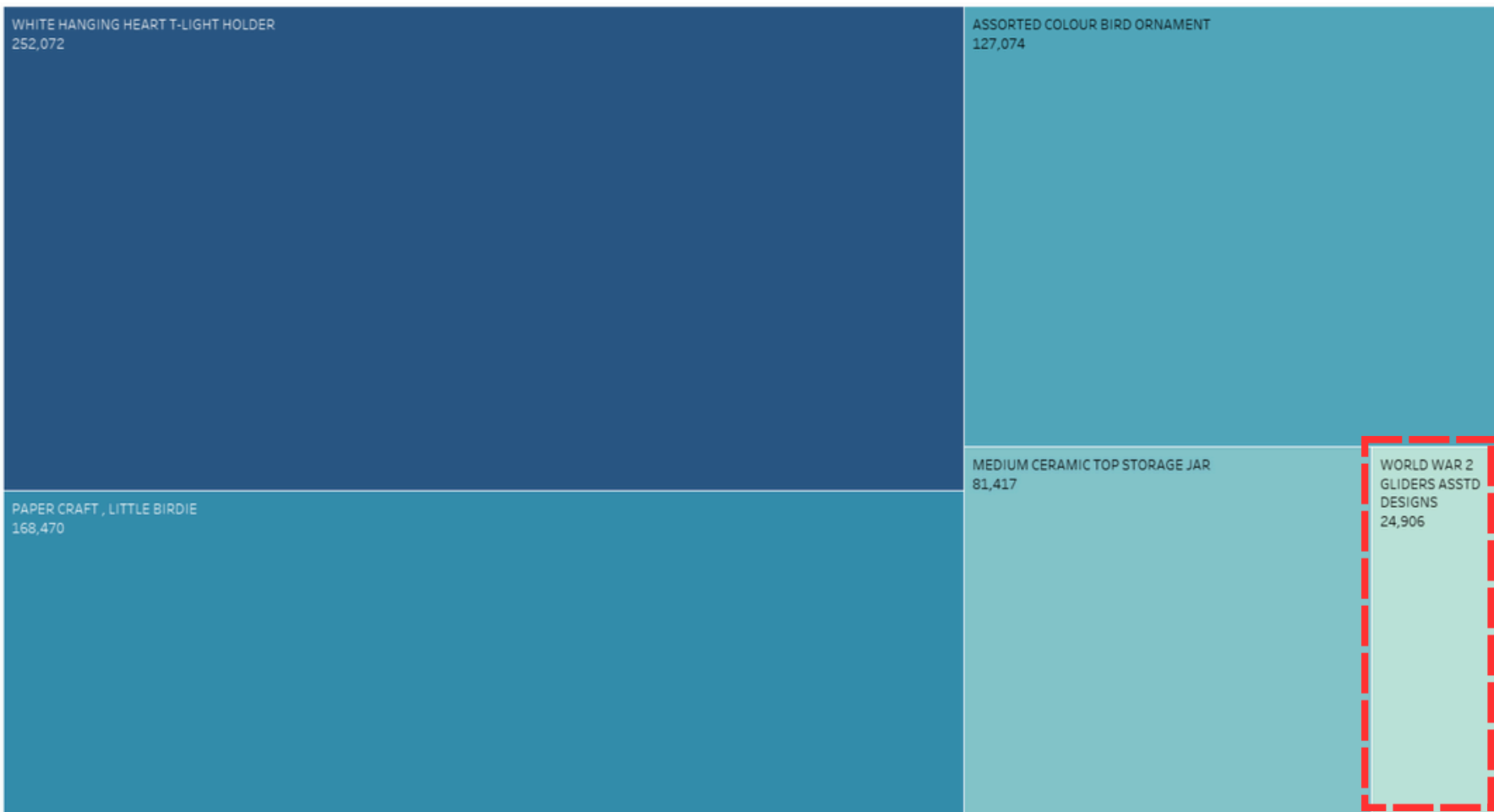
1. The online retail shop is experiencing a positive trend over the past 2 years reaching their peak in 2010 Q4
2. This trend is reflected not only for transaction but also sales wise
3. There happens to be a positive seasonal trend towards Q4 of each YEAR

Best Selling Products

Top 5 Best Selling Products by Transaction



Top 5 Best Selling Products by Sales



Key Insights

- 1. Out of the top 5 best selling items, those who perform well based on transactions also score a significant amount in sales
- 2. Despite World War 2 Gliders having the highest amount of item sold, it still contributes the lowest amount of sales among the top 5 items

RFM Analysis

	A-Z invoice	A-Z stockcode	A-Z description	123 quantity	🕒 invoicedate	123 price	A-Z customer_id	A-Z country	123 revenue	A-Z rfm_segment
1	581482	22138	BAKING SET 9 PIECE RETROSPOT	24	2011-12-09	4.25	17490	United Kingdom	102	Potential Customers
2	581482	23300	GARDENERS KNEELING PAD CUP OF TEA	48	2011-12-09	1.45	17490	United Kingdom	69.6	Potential Customers
3	581482	23251	VINTAGE RED ENAMEL TRIM MUG	96	2011-12-09	1.04	17490	United Kingdom	99.84	Potential Customers
4	581482	21875	KINGS CHOICE MUG	36	2011-12-09	1.06	17490	United Kingdom	38.16	Potential Customers
5	581482	22371	AIRLINE BAG VINTAGE TOKYO 78	12	2011-12-09	3.95	17490	United Kingdom	47.4	Potential Customers

RFM Analysis is an analysis that comprises of three different elements, namely Recency, Frequency, and Monetary.

Recency - How recent a customer makes another purchase after the date of last transaction

Frequency - The number of times a customer has made transaction

Monetary - Payment value of the transaction

Based on these 3 values, each of the data points will then be categorized into one of the 4 statistical quartiles accordingly so that segmentation analytics can be done

Our dataset will follow these criterias as means to label our segment:

- Best Customer (111)
- Potential Customers (112, 122, 211, 222)
- Lost Cheap (444, 443, 434)
- Big Spender (XX1)
- Loyal Customers (X1X)
- Others/Recent Shopper (1XX & 2XX)Almost Lost (3XX)
- Lost Customers (4XX)

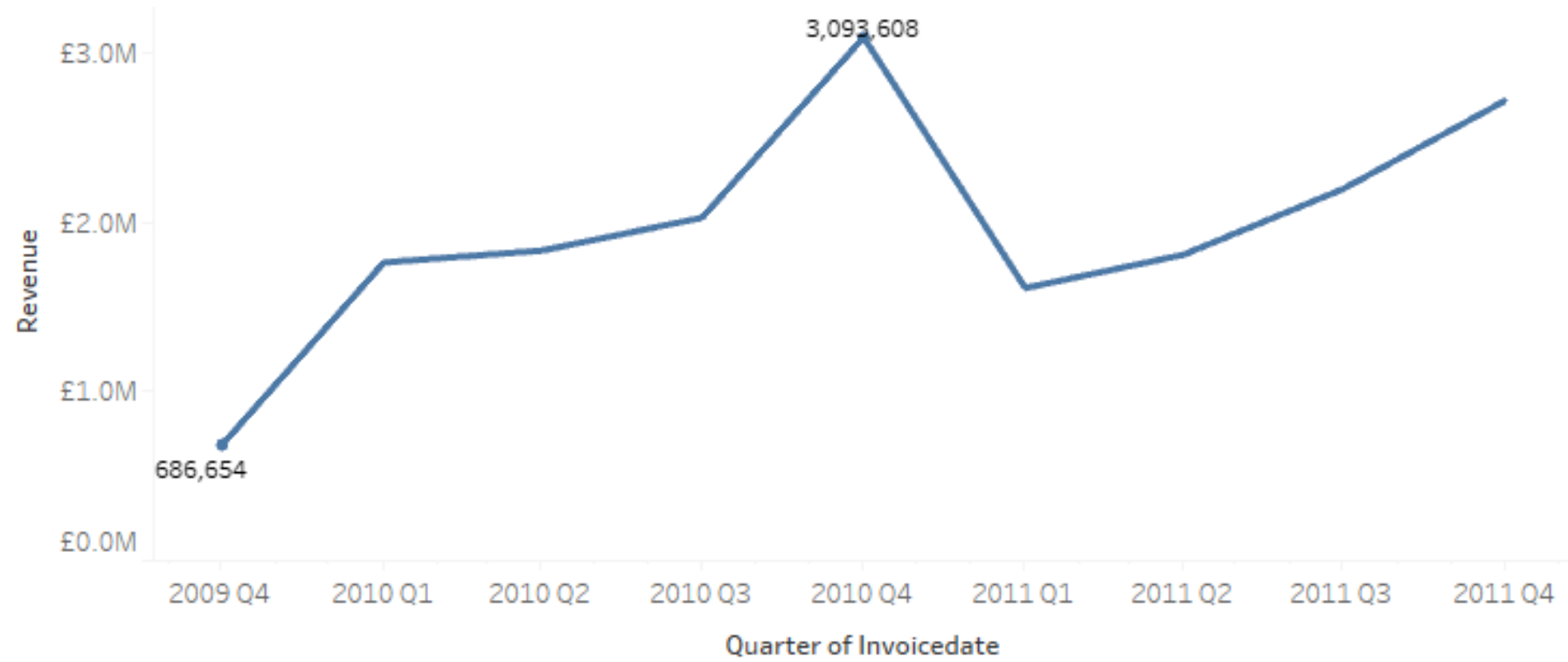
UK Retail Executive Dashboard

Total Customers
4,220
-0.31% ▼

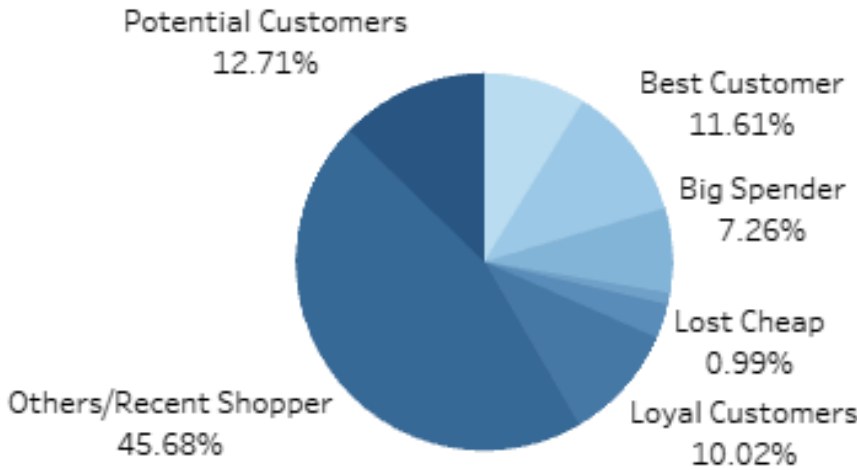
Total Sales
8,338,712
-4.35% ▼

Total Volume
4,869,417
-10.67% ▼

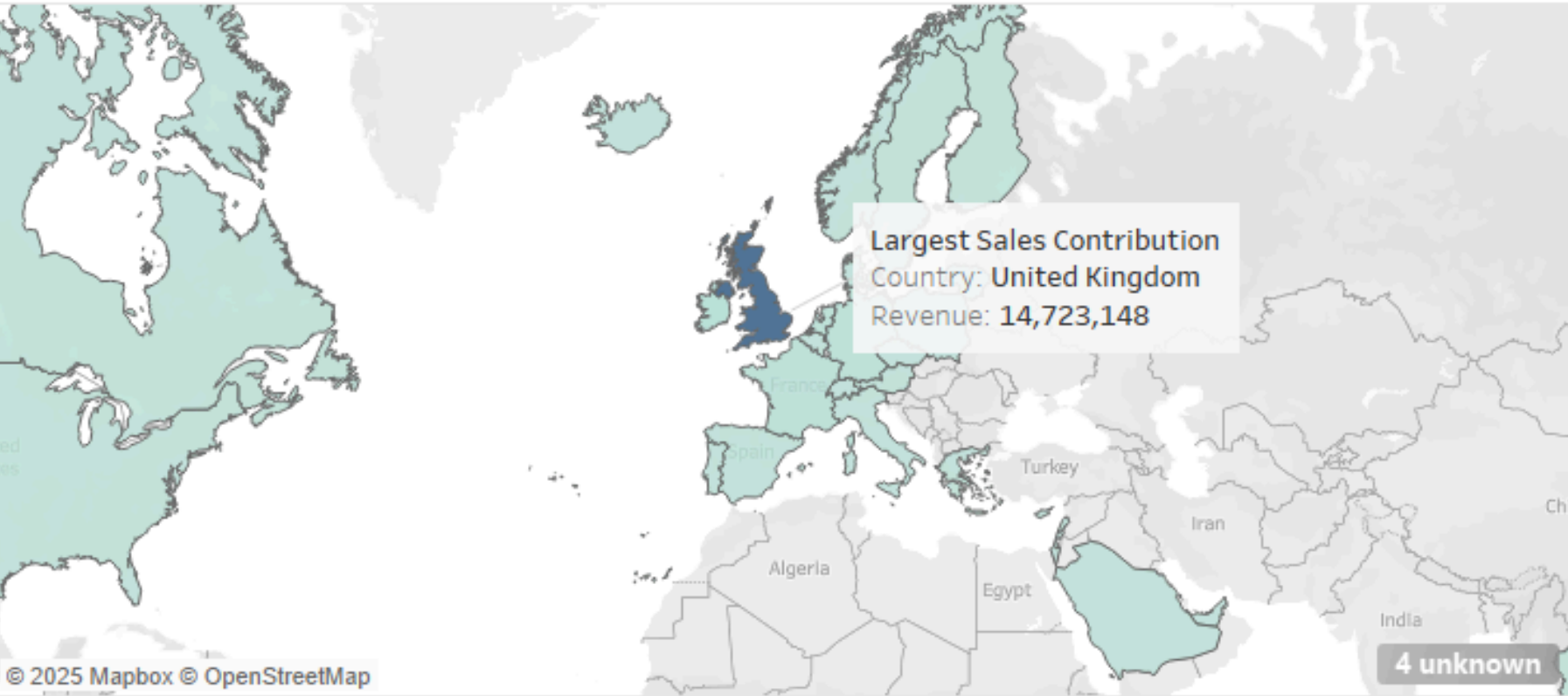
Sales Trend



RFM Segments



Customer's Geographic Contribution



Top 5 Best Selling Products



Cohort Analysis

Month of Cohor..		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1	December 2009	100.00%	37.51%	34.26%	42.78%	39.23%	39.04%	39.04%	35.79%	33.97%	37.51%	43.25%	49.57%	37.03%	29.19%	25.55%	30.53%	26.12%	30.24%	29.00%	27.46%	26.22%	31.77%	30.53%	40.86%	20.86%
	January 2010	100.00%	21.83%	30.20%	30.46%	27.92%	29.19%	26.65%	23.10%	28.93%	34.01%	30.96%	16.50%	20.56%	16.75%	18.02%	15.23%	21.83%	18.78%	17.51%	18.53%	23.60%	18.53%	22.34%	7.36%	
	February 2010	100.00%	30.03%	22.59%	30.30%	25.62%	20.94%	21.76%	28.37%	27.55%	29.20%	13.22%	12.95%	15.15%	19.28%	13.50%	20.39%	18.46%	16.80%	14.60%	23.42%	24.79%	17.08%	6.34%		
	March 2010	100.00%	21.79%	25.92%	23.62%	22.94%	19.95%	24.08%	29.82%	28.90%	11.47%	11.47%	14.22%	19.27%	16.51%	19.04%	16.97%	17.43%	15.83%	16.97%	20.41%	21.33%	7.57%			3
	April 2010	100.00%	23.02%	19.93%	16.15%	18.56%	23.02%	27.15%	26.12%	11.34%	11.68%	8.25%	14.43%	13.75%	16.15%	15.81%	14.78%	14.09%	14.09%	17.18%	20.96%	6.53%				
	May 2010	100.00%	19.29%	17.72%	19.29%	18.90%	25.98%	22.05%	12.99%	6.69%	8.66%	12.20%	13.78%	14.96%	16.14%	10.63%	12.99%	14.17%	16.54%	15.75%	4.72%					
	June 2010	100.00%	21.56%	19.70%	20.45%	23.05%	28.25%	13.01%	9.29%	8.18%	11.90%	10.41%	14.13%	15.61%	13.01%	11.15%	12.27%	13.75%	20.45%	5.95%						
	July 2010	100.00%	20.77%	20.22%	28.42%	30.05%	15.30%	11.48%	15.30%	14.21%	12.02%	13.66%	15.85%	13.66%	14.21%	20.22%	17.49%	24.59%	9.29%							
	August 2010	100.00%	24.68%	31.65%	32.28%	18.35%	13.29%	10.13%	13.92%	14.56%	13.29%	13.92%	12.66%	17.09%	20.25%	18.35%	20.25%	6.96%								
	September 2010	100.00%	30.17%	26.03%	14.05%	9.09%	10.33%	14.05%	9.92%	12.40%	14.05%	12.81%	10.33%	21.49%	17.77%	17.77%	5.37%									
	October 2010	100.00%	29.55%	16.09%	12.93%	9.23%	8.97%	13.46%	14.25%	11.35%	9.23%	10.82%	12.93%	19.26%	23.22%	7.92%										
	November 2010	100.00%	21.12%	9.63%	10.56%	9.32%	9.32%	13.66%	10.87%	8.70%	9.01%	11.49%	14.91%	26.40%	7.76%											
	December 2010	100.00%	11.69%	5.19%	9.09%	11.69%	7.79%	5.19%	9.09%	6.49%	2.60%	9.09%	19.48%	3.90%												
	January 2011	100.00%	21.13%	21.13%	19.72%	21.13%	15.49%	15.49%	12.68%	11.27%	23.94%	25.35%	11.27%													
	February 2011	100.00%	20.33%	17.07%	18.70%	21.95%	15.45%	17.07%	15.45%	16.26%	19.51%	8.13%														
	March 2011	100.00%	24.16%	24.16%	22.47%	22.47%	16.85%	21.91%	22.47%	25.28%	9.55%		3													
	April 2011	100.00%	26.67%	20.95%	20.00%	20.00%	24.76%	20.00%	25.71%	12.38%																
	May 2011	100.00%	26.85%	25.00%	17.59%	22.22%	21.30%	25.93%	11.11%																	
	June 2011	100.00%	26.85%	22.22%	27.78%	22.22%	28.70%	10.19%																		
	July 2011	100.00%	23.53%	30.39%	29.41%	35.29%	15.69%																			
	August 2011	100.00%	33.02%	31.13%	27.36%	18.87%																				
	September 2011	100.00%	34.22%	39.04%	16.04%																					
	October 2011	100.00%	33.48%	17.65%																						
	November 2011	100.00%	17.19%																							
	December 2011	100.00%																								

Key Insights

- December 2009 cohort performance shows exceptional long-term retention maintaining 20-40% retention rates even after 24 months.
- In most cases, the retention rate can hold up substantially well up to the first 6-10 months after the customer's first purchase
- There seems to be a seasonal trend going on indicating that holiday shoppers may become more loyal customers

Key Insights

- 1. December 2009 cohort performance shows exceptional long-term retention maintaining 20-40% retention rates even after 24 months.
- 2. In most cases, the retention rate can hold up substantially well up to the first 6-10 months after the customer's first purchase
- 3. There seems to be a seasonal trend going on indicating that holiday shoppers may become more loyal customers

Insights



Emerging Markets

Most of the customers are local UK residents, indicating a strong local demographic presence. Hence there may be a need to tap in to other market demographic

Positive Trend

Over the past two years, the online shop continues to generate more sales especially with the hike nearing Q4 of each year

Annual Performance

Despite experiencing a strong positive trend, 2011's performance had a 4-10% decline in sales and volume compared to 2010

Retention Rates

The first 6-10 months after the customer's first purchase tends to be crucial in making sure they return to make another purchase

Recommendations



1

Tapping into emerging markets

The platform can start using advertisements to create buzz within the EIRE and Netherland region to slowly penetrate the market

2

Loyalty Program

To make sure customers return, these shops can implement a loyalty program in which they can receive a certain amount of points after each transaction that can then be exchanged for vouchers or additional discounts for their next purchase within a fixed period of time

3

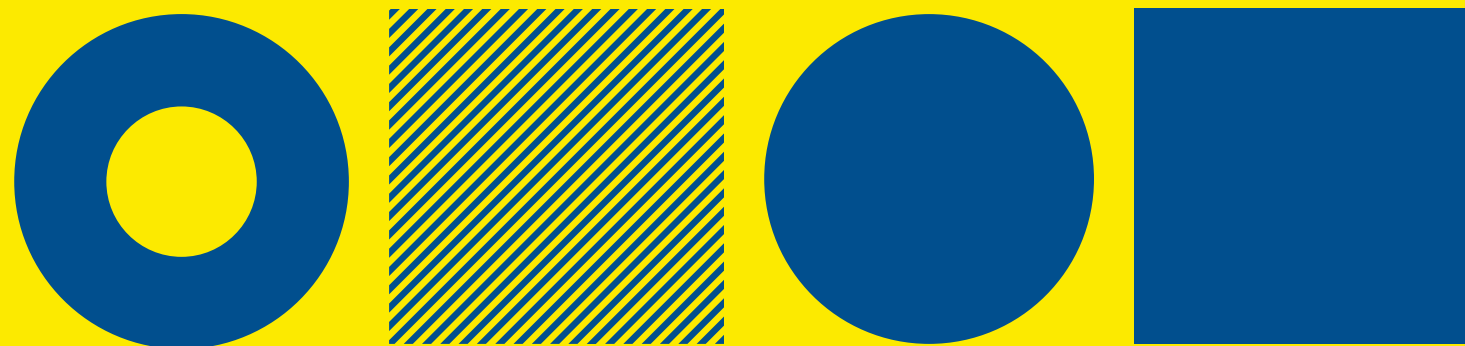
Seasonal Promotions

As the hike in sales and transactions occur during the holiday period especially nearing Q4, the shop can implement exclusive collaborated products and huge discounts within that period of time

4

Targeted Campaign & Newsletter

Since the first 6-10 months are crucial in retaining the customers from churning, with the loyalty program proposed, the company could get access to their contacts and send promotional campaigns and newsletter as a way to remind the customers to come back



Thank you